

Octavio Mordales
Lab 7

10/21/2022

I pledge my honor that I have abided by the
Stevens Honor System. Am

1. $T(n) = T(n/2) + n^2$ $a=1, b=2, k=2$ $b^k = 2^2 = 4$
Complexity: $O(n^2)$ $a < b^k$

2. $T(n) = 4T(n/2) + n^2$ $a=4, b=2, k=2$ $b^k = 2^2 = 4$
Complexity: $O(n^2 \log_2 n)$ $a = b^k$

3. $T(n) = 3T(n/3) + n$ $a=3, b=3, k=1/2$ $b^k = \sqrt{3}$ $n^{\log_3 3} = n$
Complexity: $O(n)$ $a > b^k$ $\log_3 3 = 1$

4. Recurrence: $T(n) = 2T(n/2) + n$ $a=2, b=2, k=1$ $b^k = 2^1 = 2$
Complexity: $O(n \log_2 n)$ $a = b^k$

5. Recurrence: $T(n) = T(n/2) + n^2$ $a=1, b=2, k=2$ $b^k = 2^2 = 4$
Complexity: $O(n^2)$ $a < b^k$